Student Worksheet: *Energy Transformations*

1. In a portable stereo, **electrical** energy is changed to **sound** energy.

2. When you burn wood in a campfire, **chemical** energy is converted to **heat** and **light** energy.

3. In a computer monitor, **electrical** energy is transformed into **light** energy and **sound** energy.

4. Through the process of photosynthesis, **light** energy from the sun is changed to stored **chemical** energy.

5. If you place a ball at the edge of a hill, and it rolls downhill, **gravitational potential** energy is changed to **motion (kinetic)** energy.

6. In a flashlight, the **chemical** energy in batteries is changed to **electric** energy and then to **light** energy.

7. In a fan, **electric** energy is changed to **motion (kinetic)** energy.

8. When water held behind a dam is released, **gravitational (potental)** energy is converted first to **motion (kinetic)** energy.

9. At a hydropower facility, the **motion (kinetic)** energy of flowing water can be used to spin a turbine and power a generator, creating **electric** energy.

10. In a photovoltaic (solar) panel, the **light** energy of the sun is converted directly into **electric** energy.

11. In a car, the **chemical** energy in petrol is burned to create heat energy, which is converted to **motion (kinetic)** energy.

12. In a propane-powered coffee roaster, **chemical** energy is burned to create **heat** energy.

13. In a modern wind turbine, **motion (kinetic)** energy is converted to **electric** energy.

14. A stove uses electric energy or **chemical** energy to create **heat** energy.

15. When we run, the **chemical** energy stored in our muscles is converted to **motion (kinetic)** energy.